

Designing science posters with



Yi Shuen Christine Lee

Expectation (?)

Your Poster Title Goes Here Using A Large Font Size (e.g. Arial 96)

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logo

Your Name
Affiliation



Introduction

This poster has been designed to act as a customizable template. You do not have to use this template but it might be a useful starting point. The poster includes guidance on how to change the template's background colours and fonts and import graphics as well as general advice on poster design.

Please note that the **maximum size allowed for posters at the conference is A4 size** (841x1190mm). Your poster can be smaller than A4.

What to put on your poster

- Draft a brief layout on paper first
- You should include a title, introduction, theory/methodology, findings and conclusions
- Don't just duplicate what is on your handout or in your report, try to think up other ways of presenting the information.
- Use appropriate images, diagrams and graphs to help put your message across.
- Don't forget to include your name, organisation, project title, etc and contact details.

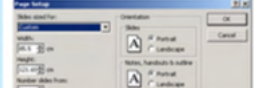
Design tips

- Keep it simple – edit hard and don't include unnecessary information (you can always have a full copy of your report handy to answer any questions).
- You should include much less text than this template!
- Keep all your columns the same width.
- Use colours sparingly. Keep to a simple scheme that is high contrast (easy to read) yet eye catching. Avoid garish or overly trendy combinations.
- Consider the capabilities of the printer when including high resolution graphics such as gradients. Some colours may print differently than they appear on screen.
- Decide on a style and stick to it – a minimum of two easy to read fonts which are compatible with the printer software (Arial, Helvetica, Times, Palatino and Garamond are best).
- As a guide body text should be minimum font size 24pt, sub-headings 30-50pt and main title 80pt+.
- Avoid using ALL CAPS or underlining to highlight a point or title. Use **bold** text instead.
- Arrange the components of your poster to appear balanced overall and maintain consistency.
- There should be an easy 'flow' through the poster for the reader. Most posters are read like a newspaper or magazine so plan your content in columns with a large banner heading.
- Avoid long numerical tables, convert into graphs or charts instead.
- Incorporate any logos where appropriate.

Creating your poster

How to use this template
Open the file in PowerPoint and click 'File', 'Save As', then give the file a new name (your poster title) and save. You can then replace any text, images, colours, fonts, backgrounds, etc with your chosen content to create your own poster. Try right clicking on items to see the options for altering backgrounds, colours, images, etc. You can easily increase or decrease the number of text boxes or add or delete images. Remember to use the 'undo' button if something goes wrong. Don't be afraid to experiment.

Getting started on your own
• Open PowerPoint, go to 'File' 'Page Set Up', enter the information in the window below and click 'OK' (or see Printing section for different size posters).



- Insert text boxes to define your columns and title bar.
- Add in formatting using the options under 'Format'. This allows you to select fonts, colours, backgrounds or to apply a theme to the whole poster.
- Remember that you are basically creating a giant PowerPoint slide so all of the formatting commands are the same.

Importing text, images and graphics
• Text can be imported from any other file (Word, e-mail, etc) using the normal highlight copy and paste commands. You may need to adjust the formatting or text box size once you've pasted the content.
• Images can be copied and pasted in the same way or inserted using the 'Insert' 'Image' commands from the toolbar. Check the resolution is high enough to cope with being enlarged (increase the zoom view to get an idea on results).

What can go wrong

A lot of poster problems can be avoided by planning ahead, so try to be organised and allow plenty of time. Being aware of the following pitfalls may help to smooth the process:

- Page size and PowerPoint** – Changing the page size after you have created your poster will alter the layout dramatically and cause imported objects to go out of shape. So always do this before you start.
- File size** – If your final document is greater than 10Mb in size there may be problems with printing. Files over 24Mb are difficult to email. You can keep a check on file size by viewing 'File' 'Properties'.
- Imported Objects** – PICT, SignalPlot, Prism, SPSS, DeltaGraph and CricketGraph files imported into PowerPoint may appear on your screen but can print incorrectly. Preferably only import JPEG or TIFF files. Imported scanned images are often problematic.
- Drafts and Proofs** – Always check the 'Scale to fit' paper' box in the print dialogue window when trying to print your drafts, otherwise you will only get a tiny corner on the printout.

Getting your poster printed

Proof Read and Test
• Thoroughly spell check and proof read your poster and then ask someone else to do it for you as well.
• Test your poster by printing off a sample A4 version (check the scale to fit box on the print menu). Check it flows nicely.
• When you are sure that the poster is ready convert it into Portable Document Format (pdf). This will secure the elements of the file so they are not changed when reopened on different computers.

Converting to Portable Document Format
1. Open the file in PowerPoint
2. Click 'File', then 'Print'
3. In the Print dialogue box select Printer name 'Adobe PDF' (if you don't have this option you will need to get someone to convert it for you)
4. Check the 'scale to fit paper' box
5. Click OK
6. The 'save file as' dialogue box appears, select the folder where you wish to save the poster, then click 'OK'
7. The pdf document should then be created. Adobe Acrobat software will launch automatically and displays it for you. The poster is now ready to print at full size, probably at a specialist print shop.

Printing for collection at the conference
• You can have your poster printed out at the university if you wish. We can print posters that are A4 (841x1190mm), A1 (594x841mm) or A2 (842x594mm).
• There is a charge for printing. A0 costs £27 unamated or £40 laminated, A1 is £15 and £21 50 and A2 £10 and £14 50 respectively.
• To use this service a pdf version of the poster must be sent to n.gales@uea.ac.uk by **Tuesday 11th July** at the very latest. You should alert Natasha beforehand if you are planning to use this service. The poster will be available for pick up at the conference registration desk.

Useful resources

Web Sites
A number of useful web sites contain advice, templates, etc, some of these are listed below. An internet search for 'poster presentations' or 'poster templates' will also throw up lots of helpful sites. Look at recent conference sites as many now put poster presentations online. You may get some design tips this way. Two good websites are:

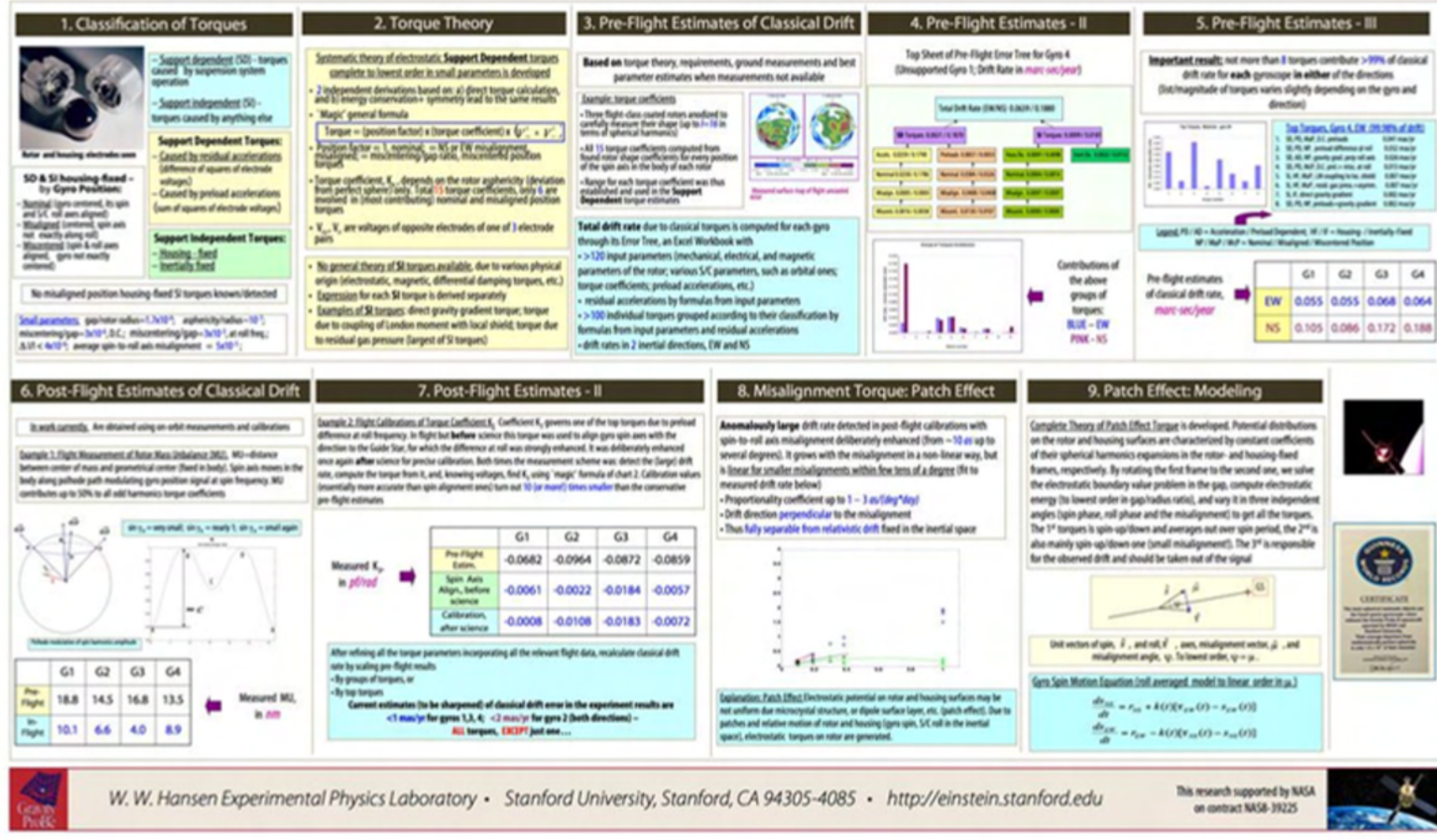
<http://www.strath.ac.uk/Departments/CAPE/poster/>
Great site for content and design ideas with example poster designs

<http://lorien.ncl.ac.uk/ukrning/Dept/Tips/present/poster.htm>
Beginner's guide to poster preparation



CLASSICAL TORQUES ON GRAVITY PROBE B GYROS

Alex Silbergleit, Mac Keiser, Yoshimi Ohshima



Identifying KIF1B as a Potential Disease Driver for Charcot Marie Tooth Disease through Phenomics-Driven Prioritization

Mikalai Malinowski, Victoria Catterston, Shahrzad Hosseini, David Kokel, Gabriel Musso
BioSymbetrics, Toronto, Ontario, Canada
MDA Conference 2023 - Poster Number: 283



Reality



Clutter...
Too much text



Too many
figures



Unclear,
roundabout
delivery

Attract, then convey

Persistent Gravitational Radiation from Glitching Pulsars

Tong Cheunchitra, Andrew Melatos, Julian Carlin, George Howitt

arXiv: 2401.05600 | LIGO DCC: 62500362

TONG CHEUNCHITRA

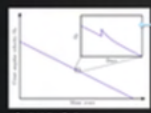
PhD Student, University of Melbourne
tcheunchitra@student.unimelb.edu.au



Superfluid vortex arrays in the core of glitching pulsars are pinned nonaxisymmetrically in-between glitches. Therefore, the superfluid emit continuous gravitational wave (GW) in-between each glitch. The scaling of the GW amplitude with frequency depends strongly on microphysics – specifically, the size of vortex pinning compared to inter-vortex repulsion.

Pulsar Glitches

- Pulsars are rapidly spinning neutron stars.
- Pulsar spin decelerate gradually over time due to electromagnetic radiation.
- However, in some pulsars, we observe **glitches**: occasional abrupt increases in spin frequency.



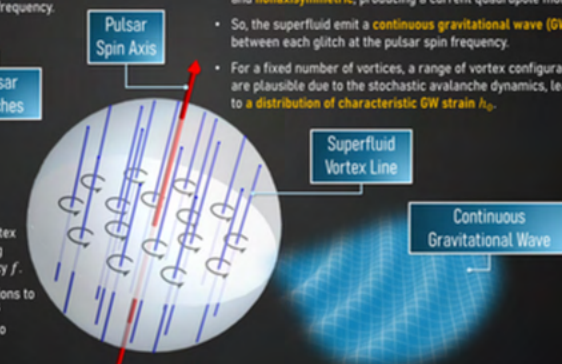
Estimating GW Strain

- We used an N-body simulation of vortex avalanches [1] to estimate the scaling between h_0 and pulsar spin frequency f .
- Computational cost restricts simulations to $\sim 10^4$ vortices, compared to $\sim 10^{15}$ vortices in real pulsars, so we need to extrapolate the scaling.

[1] Howitt et al. (2022)

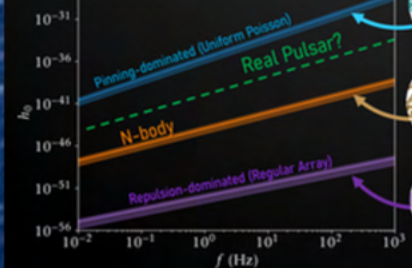
Vortex Avalanches

- Pulsar glitches may be explained by **vortex avalanches**: the sudden, collective movement of vortices in the superfluid core of the pulsar.
- The spatial configuration of vortices in-between glitches is **fixed** and **nonaxisymmetric**, producing a current quadrupole moment.
- So, the superfluid emit a **continuous gravitational wave (GW)** in-between each glitch at the pulsar spin frequency.
- For a fixed number of vortices, a range of vortex configurations are plausible due to the stochastic avalanche dynamics, leading to a **distribution of characteristic GW strain h_0** .



GW Strain Scalings

- The N-body simulation predicts a GW scaling
- $$h_0(f) = 7.3^{+7.9}_{-5.4} \times 10^{-42} \left(\frac{f}{30 \text{ Hz}}\right)^{1.9} \left(\frac{D}{1 \text{ kpc}}\right)^{-1}$$
 for a fiducial pulsar.
- To ensure that the extrapolated scaling is sensible, we develop **analytic scalings** for limiting cases in the vortex configuration/microphysics:
- The astrophysical $h_0(f)$ scaling lies somewhere in-between, depending on the size of vortex pinning compared to inter-vortex repulsion in neutron stars.



THE GRAVITATIONAL WAVE BACKGROUND



In June 2023 there was a coordinated release by the International Pulsar Timing Array of the first evidence of the gravitational wave background

What are pulsars?

Pulsars are 'dead' stars that rotate so regularly we can use them to keep time. They are as massive as the Sun, smaller than Melbourne and spin faster than a blender. Pulsars are natural cosmic clocks as accurate as atomic clocks

How can we detect gravitational waves with pulsars?

Gravitational waves are ripples in spacetime, so when a wave passes between Earth and a pulsar it changes the amount of spacetime between them. This affects the light travel time between the pulsar and Earth, leading to a change in the time the pulse arrives. We time many pulsars across the galaxy in a coordinated way to form a Pulsar Timing Array.

We have effectively created a telescope the size of the galaxy!

Why is understanding the gravitational wave background important?

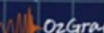
We know that huge black holes - billions of times more massive than the sun - live in the centre of galaxies, but how to form black holes that big remains a mystery! The gravitational wave background likely comes from the collisions of huge black holes, which may form the building blocks for supermassive black holes. By 'hearing' these collisions in the gravitational wave background we may be able to understand how these black holes grow!

While the gravitational wave background likely comes from colliding black holes, it could come from events in the early Universe, just after The Big Bang! The early Universe was very hot and dense, akin to a pot of boiling water. Collisions between 'bubbles' in this early Universe soup may generate gravitational waves. If the gravitational wave background does in fact come from cosmic soup shortly after the Big Bang, we will need to re-write our fundamental understanding of physics!

Ask me about

WOMEN AND NON-BINARY PEOPLE

Rowina Nathan (she/her) | School of Physics and Astronomy



@astro_taclarke teaganclarke.com teagan.clarke@monash.edu
LVK Meeting, March 2023, Northwestern University

NSBH TIDAL DISRUPTIONS: the best way to measure equation of state?

Teagan Clarke, Paul Lasky and Eric Thrane

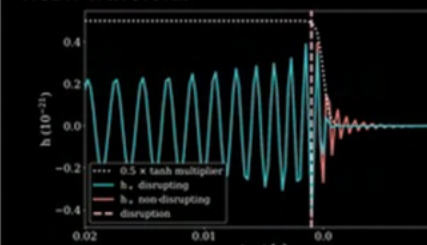


INTRO

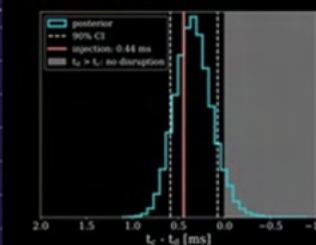
Some NSBH may undergo tidal disruption. Measuring the moment of disruption can provide information about the neutron star equation of state. We simulate this with an optimised system measured in Cosmic Explorer.

MODEL

BNS waveform x Tanh = phenomenological NSBH waveform



RESULTS



SUMMARY

- We can accurately measure the moment of tidal disruption in gravitational waves with Cosmic Explorer.
- However we gain more information about EOS from tides than from disruption time
- This can provide a consistency check between disruption and tides
- Hints of new physics? phase transitions in neutron stars?
- We still love NSBH for binary stellar evolution physics.

CHECK OUT OUR PAPER!
arXiv:2302.09711



WANTED

Sruthi Suresh
sruthi@student.unimelb.edu.au

Harry Potter and The width of the emission lines

Type 1 and Type 2 AGN Catalogue from 6dFGS in the Southern Sky
Sruthi Suresh, Wei Jent Hon, Rachel Webster, Christian Wolf, Christopher Onken

TL; DR Took 6dFGS spectra. Ran PyQSOFit. Checked the fits by eye (My eyes, my eyes!) Outcome - An extensive catalogue with 2,515 Type 1-1.9s, 5,071 Type 2 AGN.

The Restricted Library:

- Parent Sample: 136,604 6dF Galaxy survey (Jones et al. 2004, 2009)
- Location: UKST at SSO; 5.7° diameter field, covers 17,000 square degrees, 10 times larger than 2DFRS (Colless et al. 2001)
- Primarily focuses on NIR and optical
- Two grating setup - Y-band, -4000Å to -5600Å, R-band, -5600Å to 7500Å

You're an AGN, Gal!

6dFGS sources (117,610) → Clean spectrum → Statistical cut to identify emission line galaxies (MAD) → Spectra visual inspection → Fit spectrum using PyQSOFit → Filter AGN candidates → Classification (BPT - Type 2, flux ratio - Type 1) → Examine optical images → Calibrate flux and get luminosities → Broad emission lines - Type 1 (Webster et al. 2024) → Narrow emission lines - Type 2 (Suresh et al. submitted)

That this catalogue is up to ALL GOOD!

Sorting hat

- Requires [OIII] & Hα to be over (4.5 MAD)
- Find peaks height ratio is used to discard SF galaxies
- Visually inspected - 50,000 spectra to select AGN

Type 1 - Broad and bold

Classification is based on the following:

- Type-1.0 $5.0 \leq R$
- Type-1.2 $2.0 \leq R < 5.0$
- Type-1.5 $0.333 \leq R < 2.0$
- Type-1.8 $R < 0.333$

Where $R = \frac{f_{H\beta}(\text{broad})}{f_{[OIII]5007\text{Å}}}$

Identified 2,515 unique broad-line AGNs (BLAGN) with a median redshift of 0.207

Category	Count
2 < R	740
0.1 < R < 0.5	934
0.5 < R < 1.5	454 (with Maffi)
1.5 < R	207 (high redshift OSG)

Type 2 - Hidden under cloak

Classification based on BPT diagram (Baldwin et al. 1981)

- Sources between the Kewley line and the Kauffmann line are classified as Composite (COMP)
- Around 1000 sources do not have Hβ; the pseudo value calculated using Hα is used to plot BPT.
- Visually inspected optical images (PanSTARRS, DSS) of sources to remove mergers, 8,917 single uncontaminated sources found.

Category	Count
AGN	2,744
COMP	2,327
SF	5,432

Crystal ball - Future teller:

- Multi-wavelength Study - Observe in X-ray, IR and UV.
- LSST - Track changes in emission lines. Observe the time curve using photometry from LSST.
- True Type 2 - Is the Unification theory wrong?

The Author's Guide to AGN:

James D. H. et al. 2004, Monthly Notices of the Royal Astronomical Society, 355, 947
Baldwin J. A., Phillips M. M., Terlevich R. 1981, PASP 93, 5
Woff C. et al. 2014, Publ. Astron. Soc. Australia, 36, 469
Osten C. A. et al. 2024, Publ. Astron. Soc. Australia, 45, 466
Chen Y. P. et al. 2022, The Astrophysical Journal Supplement Series, 258, 19

I'm ALWAYS up for a chat! Connect with me

Personal branding necessary, but help

Enhancing gravitational-wave burst detection confidence in expanded detector networks with the *BayesWave* pipeline

Yi Shuen Christine Lee, Meg Millhouse, Andrew Melatos

PHYS. REV. D 103, 062002 (2021)

Expanded detector networks

The Laser Interferometer Gravitational-Wave Observatory (LIGO) has completed three observing runs (O1, O2, O3) with Virgo joining at the end of O2 and the whole of O3. The Kamioka Gravitational Wave Detector (KAGRA) also began observing at the end of O3. The global gravitational-wave detector network achieves higher detection rates, better parameter estimates, and more accurate sky localisation as number of detectors, $\mathcal{F}$, increases.

BayesWave: unmodelled burst search pipeline

BayesWave is a source-agnostic gravitational-wave (GW) burst detection algorithm that reconstructs non-Gaussian, transient features in the data as a sum of sine-Gaussian wavelets. Model selection in *BayesWave* is done by comparing model evidences via the Bayes factor.

Figure 1: Top panel shows individual sine-Gaussian wavelets, each of different colour, used in the reconstruction of GW150914 in the time domain. Bottom panel shows the resulting waveform (blue) as a sum of all wavelets, overlaid on the actual data (grey). [Image courtesy of Meg Millhouse]

BayesWave and Expanded Detector Networks

Aim: Quantify *BayesWave*'s detection confidence as a function of number of detectors, $\mathcal{F}$. We use the Bayes factor, $\mathcal{B}_{\mathcal{F},\mathcal{S}}$ between the Gaussian-noise plus signal model ($\mathcal{S}$) and the Gaussian noise plus glitch model ($\mathcal{G}$) to measure detection confidence. The three configurations used in this study are: LIGO Hanford-Livingston, (HL), HL-Virgo (HLV) and HLV-KAGRA (HLKV).

Analytic Bayes factor scaling

We use Laplace approximation to estimate the Bayesian evidence and hence the primary scaling of $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ [3]. We find that $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ scales mainly with the $\mathcal{F}$, number of wavelets (N) and signal-to-noise ratio (SNR_{net}) of the injected waveform:

$$\ln \mathcal{B}_{\mathcal{F},\mathcal{S}} \sim \mathcal{F} N \ln \text{SNR}_{\text{net}}$$

Empirical Bayes factor scaling

Method: Inject 150 phenomenological binary black hole (BBH) waveforms into simulated Gaussian noise coloured by the projected power spectral density (PSD) of LIGO, Virgo and KAGRA for the fourth observing run (O4). We then use *BayesWave* to recover the injections to obtain empirical scaling of $\mathcal{B}_{\mathcal{F},\mathcal{S}}$. The BBHs have equal component masses of $30M_{\odot}$ and are modelled using IMRPhenomD [2].

Results & discussion: Figure 2 shows $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ as a function of SNR_{net} for BBH injections recovered using *BayesWave*. All three networks show increasing $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ with increasing SNR_{net} . However, injections at comparable SNRs are recovered with higher $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ in the larger networks. To test the scaling of $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ with $\mathcal{F}$ alone, we use *BayesWave* to recover a set of ad-hoc sine-Gaussian wavelets ($N = 1$) injected as coherent signals from the three configurations. Figure 3 shows agreement between the analytic and empirical results.

Figure 2: $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ of BBH injection recoveries versus SNR_{net} . Each data point represents a single BBH injection.

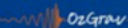
Figure 3: $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$ of sine-Gaussian wavelet injection recoveries versus SNR_{net} . The solid lines with colors corresponding to the data symbols are analytic predictions of $\ln \mathcal{B}_{\mathcal{F},\mathcal{S}}$.

Acknowledgement & References

This material is based upon work supported by the LIGO Laboratory fully funded by the National Science Foundation. Poster background © Mark Myers (OzGrav).
 [1] N. Cornish, T. Littenberg, *Classical Quantum Gravity* 32, 135012 (2015).
 [2] S. Husa et al., *Phys. Rev. D* 93, 044006 (2016).
 [3] T. Littenberg et al., *Phys. Rev. D* 94, 044050 (2016).
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 Twitter: @AstroOrca



THE UNIVERSITY OF
MELBOURNE



O3GRAV

arXiv:2008.01234 [astro-ph.HE]

Gravitational-wave burst detection efficiency of the BayesWave pipeline with Multi-detector networks

Yi Shuan Christine Lee, Meg Millhouse, Andrew Melatos

INTRODUCTION

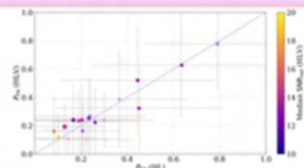
Detection confidence of the source-agnostic gravitational-wave (GW) burst search pipeline *BayesWave* [1] is quantified by the log signal-to-noise-glitch Bayes factor, $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$. A recent study [2] shows that $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$ increases with the number of detectors. However, as the detector network expands, non-Gaussian detector events (glitches) become more frequent. Glitches can mimic or mask burst signals resulting in false alarm detections, consequently reducing detection significance. This work presents an empirical study on the **impact of increased false alarm probability on the overall performance of *BayesWave* with expanded detector networks**, comparing between the Hanford-Livingston (HL, two-detector) and Hanford-Livingston-Virgo (HLV, three-detector) networks.

NOISE BACKGROUND

False alarm probability (P_{FA}) is the probability that an event detected with some $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$ is a false alarm. P_{FA} is used to quantify *BayesWave*'s detection significance, where low P_{FA} indicates high astrophysical significance. P_{FA} is measured using non-astrophysical background data, down-selected from coherent-noise bursts (CWBs) trigger list for first half of Advanced LIGO-Virgo Third Observing Run (O3) [3, 5]. Assuming the detector noise background can be modelled as a Poisson process, the shaded regions show the 1- σ Poisson uncertainty regions. The background measurements show that P_{FA} is higher for HLV than for HL, at all $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$ because the occurrence of background triggers increases with the number of detectors. As a result, events detected by the HL network need to attain a higher $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$ to achieve the same significance as the HL network.

EFFICIENCY CURVES

Detection efficiency ($\mathcal{P}_{DE}$) of *BayesWave* is measured as the probability of detecting an astrophysical event at a given significance. We combine $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$ of simulated binary black hole (BBH) signals with the background measurements to construct **efficiency curves**, which plots $\mathcal{P}_{DE}$ against P_{FA} . By comparing the efficiency curves within the astrophysical-relevant region ($\ln \mathcal{B}_{\mathcal{L},\mathcal{G}} \leq 0.4$), we find that BBH overall detection efficiency is similar for HL and HLV.



• GW150808_155046	• GW190521	• GW190521_225702
• GW180322	• GW190612_174927	• GW190701_023846
• GW180323_183804	• GW190716_220451	• GW200126_005438
• GW180324_180714	• GW190727_080323	• GW200202_154153
• GW190513_205428	• GW190728_050805	• GW200511_131703
• GW190519_155544	• GW190828_080305	• GW200224_233214

SIGNIFICANCE OF O3 GW CANDIDATES

As a consistency check, we analyse a set of O3-like compact binary coalescence (CBC) signals i.e. off-source waveforms with parameters sampled from the match-filter posterior of O3 GW detection events [4, 5]. We use the background measurements above to quantify the significance of 18 GW events, each with their own set of off-source waveforms. The figure to the left compares the median P_{FA} between HL and HLV off-source waveforms. Events with higher network signal-to-noise ratios are detected with lower P_{FA} , but all 18 events lie close to the diagonal line where $P_{FA}(\text{HL}) = P_{FA}(\text{HLV})$. This observation supports that *BayesWave*'s recovery rates with similar significance for both HL and HLV.

CONCLUSION

The results from two independent analyses show that there are no major differences between *BayesWave*'s overall performance with the HL and HLV networks. This is because the increased false alarm probability (P_{FA}) in larger detector networks offsets the advantage of increased $\ln \mathcal{B}_{\mathcal{L},\mathcal{G}}$. Our findings are consistent with previous studies [4, 7].

This material is based upon work supported by the LIGO Laboratory fully funded by the National Science Foundation. We acknowledge the use of the A3 background trigger lists.

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Twitter @astrophysics

Bayesian Inference and Gravitational Waves in Undergraduate Computational Physics

Yi Shuen Christine Lee¹, Kok Hong Thong, Yutong Bu
School of Physics, The University of Melbourne, Victoria 3010, Australia.

PROBLEM STATEMENT

Bayesian inference is a widely used statistical tool which prevails among diverse research areas including science, engineering, health care, social science, sports etc. Despite its broad applicability in various physics experiments, Bayesian inference is often overlooked or under-discussed in undergraduate physics curriculum. Furthermore, undergraduate coursework strongly focuses on the theoretical aspect of physics with a lack of mention about engaging research work.

- We present a *research-informed computational physics lab activity* which introduces Bayesian inference in the context of **gravitational-wave (GW) detection**, a cutting-edge research topic awarded the 2017 Nobel Prize in Physics.
- This activity targets **third year undergraduate students** majoring in physics, to foster interests in pursuing further studies and research in physics.
- We discuss the structure and objectives of this activity, while showcasing the results produced by students who have undertaken the activity.

BAYESIAN INFERENCE

KEY CONCEPT

Bayes' Theorem as a tool to update probability distributions of hypotheses using existing knowledge and new observations.

Bayes factor as a tool for comparing two competing statistical models for a given dataset

Markov Chain Monte Carlo (MCMC) methods as a numerical sampling tool

ACTIVITY

A medical test example

Using Bayes' theorem to calculate the probability of obtaining a true positive COVID RAT result when tested on a randomly selected individual

A model selection example

Using the Bayes factor to compare of two velocity probability distributions, modelled by Maxwell-Boltzmann (MB) equation, is more strongly supported by a given dataset

Students write a Metropolis-Hastings sampling algorithm in the Python programming language to estimate the parameter of a model for a given dataset

[Left] Students are provided with an arbitrary dataset containing a range of velocities, v .

[Right] Assuming dataset can be modelled by the MB equation, parameterised by the peak velocity v_0 . Students write their own Metropolis-Hastings algorithm in Python to sample the posterior distribution of v_0 .

GRAVITATIONAL-WAVE DATA ANALYSIS

With the foundations of Bayesian inference in hand, students now work with the data of a Bayesian GW detection algorithm called **BayesWave [1]**. **BayesWave** is one of the leading transient GW search algorithms in the LIGO-Virgo-KAGRA collaboration. The aim of this exercise is to give students the experience of managing, analysing and presenting large data sets. Students choose between one of the two following activities.

WAVEFORM RECOVERY

This exercise simulates the experience of detecting a GW while introducing the concept of **credible intervals** (CI) in Bayesian statistics.

Students extract relevant output from the **BayesWave** analysis to visualise GW signals. The **plot below** shows: (i) the detector data (red) containing the signal of GW150914, the first ever GW detected in 2015 and (ii) the 90% credible interval of **BayesWave**'s representation of the signal (blue).

SIGNAL PROCESSING

This exercise demonstrates noise whitening, a signal processing procedure applicable beyond GW searches.

Noise power in GW detectors fluctuates across the frequency spectrum (i.e. noise is coloured). The whitening process equalises the noise power, so that excess signal power at any frequency becomes more obvious. The left panel below shows a coloured GW signal. The **right panel** shows the same signal **whitened** by a student (blue), compared to the signal whitened by **BayesWave** (orange).

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THE UNIVERSITY OF
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OSCAR

Characterization of gravitational-wave burst polarizations with the BayesWave algorithm

Yi Shuen C. Lee, Meg Millhouse, Andrew Melatos



OSCAR logo

OSCAR logo



The global gravitational-wave (GW) detector network currently consists of four interferometers: LIGO Hanford [6] and Livingston [5], Virgo [6] and KAGRA [8], the commissioning of UGO India is also well underway. The expanding detector network enables more accurate probing of GW polarizations, as each detector measures the combination of polarization states independently. Bayesian methods [1] work as a source-agnostic analysis pipeline designed for the joint detection and characterization of GW transients (a. bursts) and inspirals [2]. In this work, we present a multi-detector analysis on BayesWave's ability to (i) recover polarization content and (ii) reconstruct and distinguish between GW signals from precessing and non-precessing binaries.

BAYESWAVE RECONSTRUCTION

BayesWave reconstructs transient, non-Gaussian features in the data as a sum of n individual Gaussian waveforms. Signal models in earlier versions of BayesWave are restricted by signal polarizations [3]:

$$h_i = \sum_{\alpha} \sqrt{A_{\alpha}} \cos(\omega t + \phi_{\alpha} + \alpha) \quad (1)$$

where α encodes the degree of polarization, ∇ denotes time domain quadrature, n is the total number of wavelets, and the vector is parametrized by: central time t_0 , central frequency f_0 , quality factor Q , amplitude A and phase ϕ . In a recent update [3], BayesWave releases the algorithm constraints to optimize the characterization of complex signals like white noise bursts and high-frequency supernova. This **relaxed polarization model** (3) allows α , and α can be reconstructed independently while assuming the same ω , ϕ , A , Q and t_0 for both polarization states [3].

NEW SIGNAL PARAMETERS

Polarization content of a monochromatic plane wave is typically encoded in terms of Stokes parameters, for polychromatic GW signals, the Stokes parameters are a function of frequency. BayesWave assumes purely tensorial polarizations as predicted by GR, while also takes the SSBs parameters are given by [3]:

$$I = \frac{1}{2} (h_+^2 + h_{\times}^2), \quad Q = \frac{1}{2} (h_+^2 - h_{\times}^2), \quad U = h_+ h_{\times}, \quad V = h_+ h_{\times}^2$$

As in electromagnetism, I describes the total intensity, $Q = \sqrt{Q^2 + U^2 + V^2}$ measures the degree of linear polarization, the preference of h_+ over $h_{\times}$, U and V measures the degree of circular polarization and $V = \sqrt{U^2 + V^2}$ measures the total polarization of the signal as expected to gravitational.

INJECTION TESTS

- **Melatos+PhD** - Phenomenological linear black hole (BBH)
- **30 events**, Component mass: $40 - 100 M_{\odot}$, primary black hole, Final initial inclination of 40°
- **Set (1)** Zero spin, non-precessing and **Set (2)** spin-aligned, **precessing** binaries, where the initial spin for each component mass is independently sampled within $0.1 \leq |i| \leq 1$ [3] ± 0.1

BAYES FACTOR - (ELLIPTICAL (E)) vs. RELAXED (R)

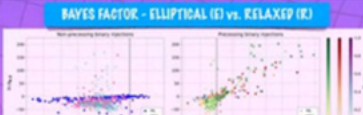


Figure 1: Bayes factor between R and E models. The plot shows log10(Bayes factor) vs. log10(SNR). The x-axis ranges from -2 to 2, and the y-axis ranges from -4 to 4. Data points are colored by SNR: blue for SNR < 10, green for 10 < SNR < 20, yellow for 20 < SNR < 30, orange for 30 < SNR < 40, red for 40 < SNR < 50, and purple for SNR > 50. A vertical line at log10(SNR) = 0 indicates equal evidence. The plot shows that for SNR > 10, the relaxed model (R) is preferred over the elliptical model (E).

STOKES PARAMETER RECOVERY

We used the sum of squared errors (SSE) to quantify the discrepancy between the injected (id) and recovered (id) polarization functions (E, R, T) in the frequency domain. The SSE is defined as:

$$SSE = \sum_{\alpha} \left[\frac{1}{2} (h_{id}^2 - h_{id}^2) \right]$$

To optimally assess BayesWave's ability in recovering polarization content, we set the frequency range to $f_{min} = 40$ Hz and $f_{max} = 400$ Hz. Our detector noise floors are lowest. In Fig. 2 we compare the SSE of the HL and HVC network against HVC for the precessing binary injections. The SSE for H, injections (in purple) are generally higher than HVC for all polarization functions, suggesting that H, recovers the polarization content less accurately than HVC. The SSE for R, injections (in orange) are generally lower than HVC for all polarization functions, suggesting that R, recovers the polarization content more accurately than HVC. The SSE for T, injections (in green) are generally higher than HVC for all polarization functions, suggesting that T, recovers the polarization content less accurately than HVC. The SSE for E, injections (in blue) are generally higher than HVC for all polarization functions, suggesting that E, recovers the polarization content less accurately than HVC. The SSE for R, injections (in orange) are generally lower than HVC for all polarization functions, suggesting that R, recovers the polarization content more accurately than HVC. The SSE for T, injections (in green) are generally higher than HVC for all polarization functions, suggesting that T, recovers the polarization content less accurately than HVC. The SSE for E, injections (in blue) are generally higher than HVC for all polarization functions, suggesting that E, recovers the polarization content less accurately than HVC.

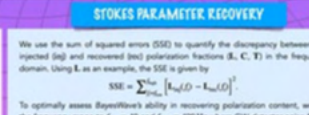


Figure 2: SSE of injected and recovered Stokes parameters. The plot shows log10(SSE) vs. log10(SNR). The x-axis ranges from -2 to 2, and the y-axis ranges from -4 to 4. Data points are colored by SNR: blue for SNR < 10, green for 10 < SNR < 20, yellow for 20 < SNR < 30, orange for 30 < SNR < 40, red for 40 < SNR < 50, and purple for SNR > 50. A vertical line at log10(SNR) = 0 indicates equal evidence. The plot shows that for SNR > 10, the relaxed model (R) is preferred over the elliptical model (E).

KEY TAKEAWAYS

This research is a hard-won quest supported by the LIGO Laboratory. We thank the National Science Foundation.

We used BayesWave's ability to characterize GW burst polarizations using simulated precessing and non-precessing BBH signals. We compare the Bayes factor for the elliptical (E) and relaxed (R) models with the precessing BBHs as the output of HVC. The increasing evidence is a parameterization with larger detector networks. We also show that larger network recoveries are more accurate, using SSE as the metric. As the global detector network continues to expand, there is a potential for more effectively distinguishing between precessing and non-precessing binaries as the Bayes factor and more accurately probing the polarization content of GW signals.

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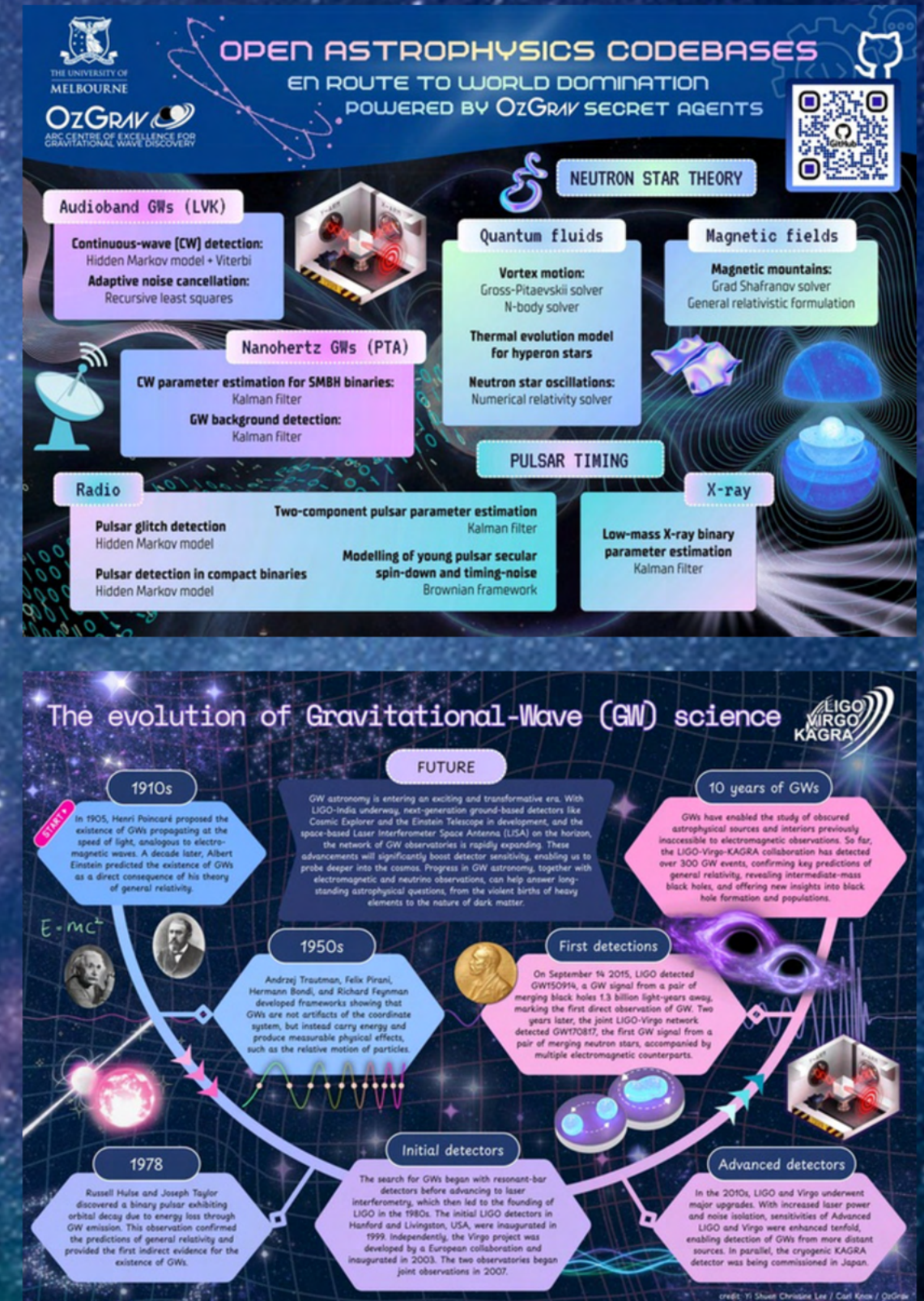
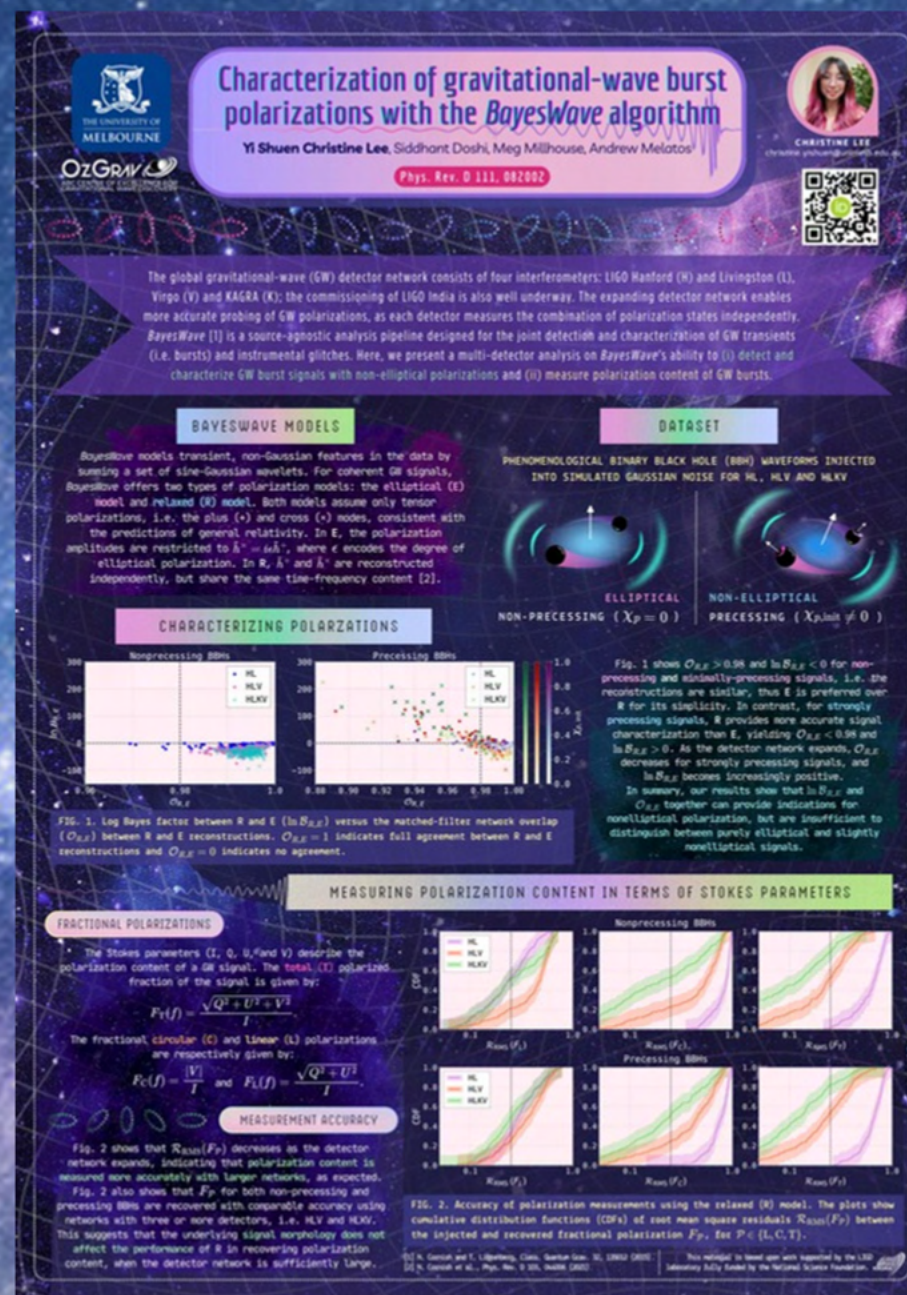
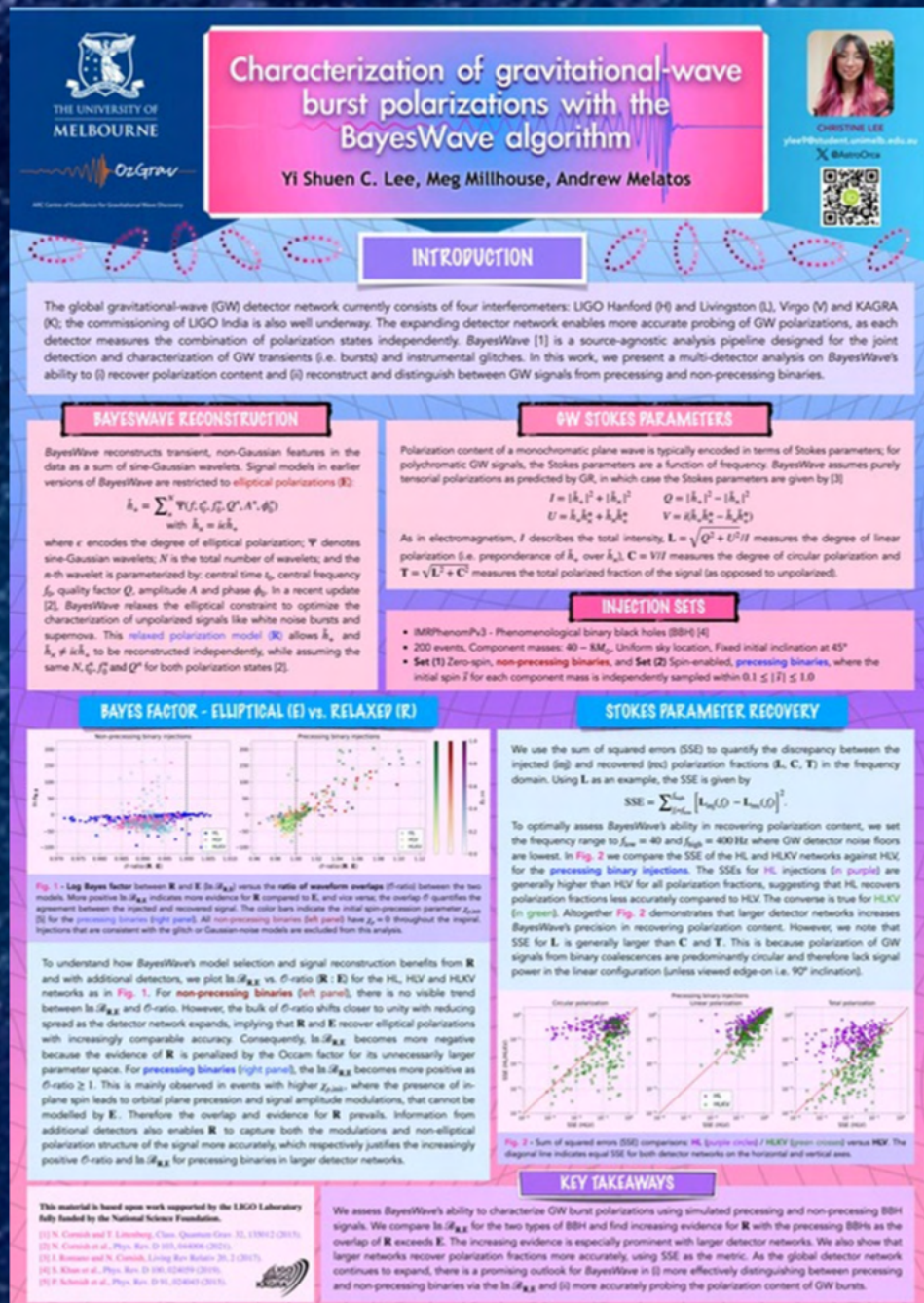
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CAUTION

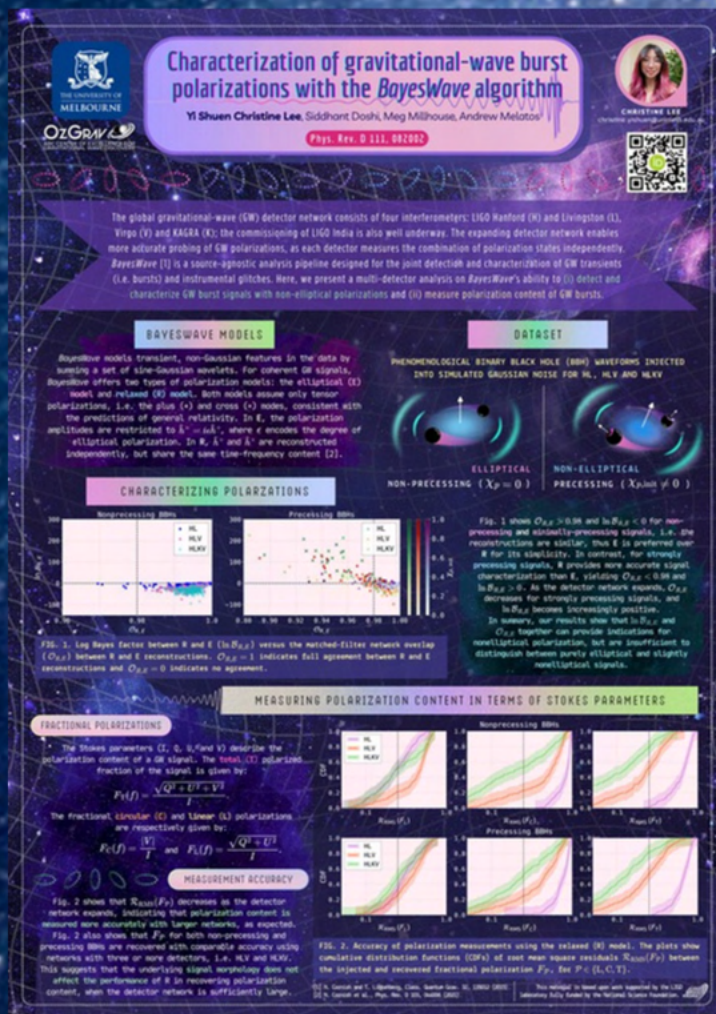
These posters are
too wordy!!

Personal branding (not necessary, but helpful!)

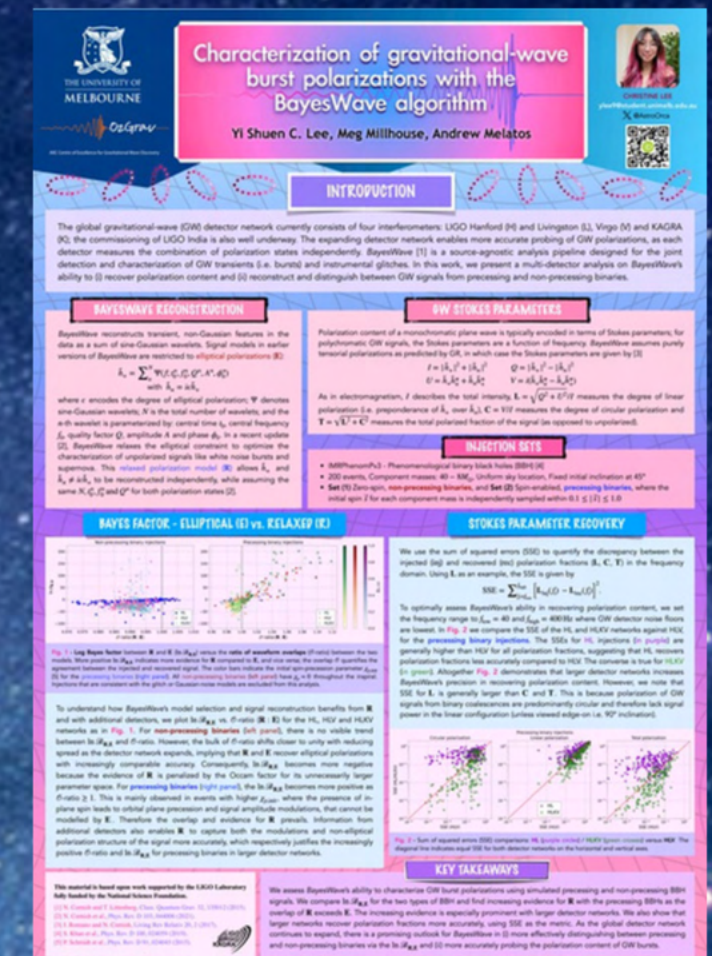
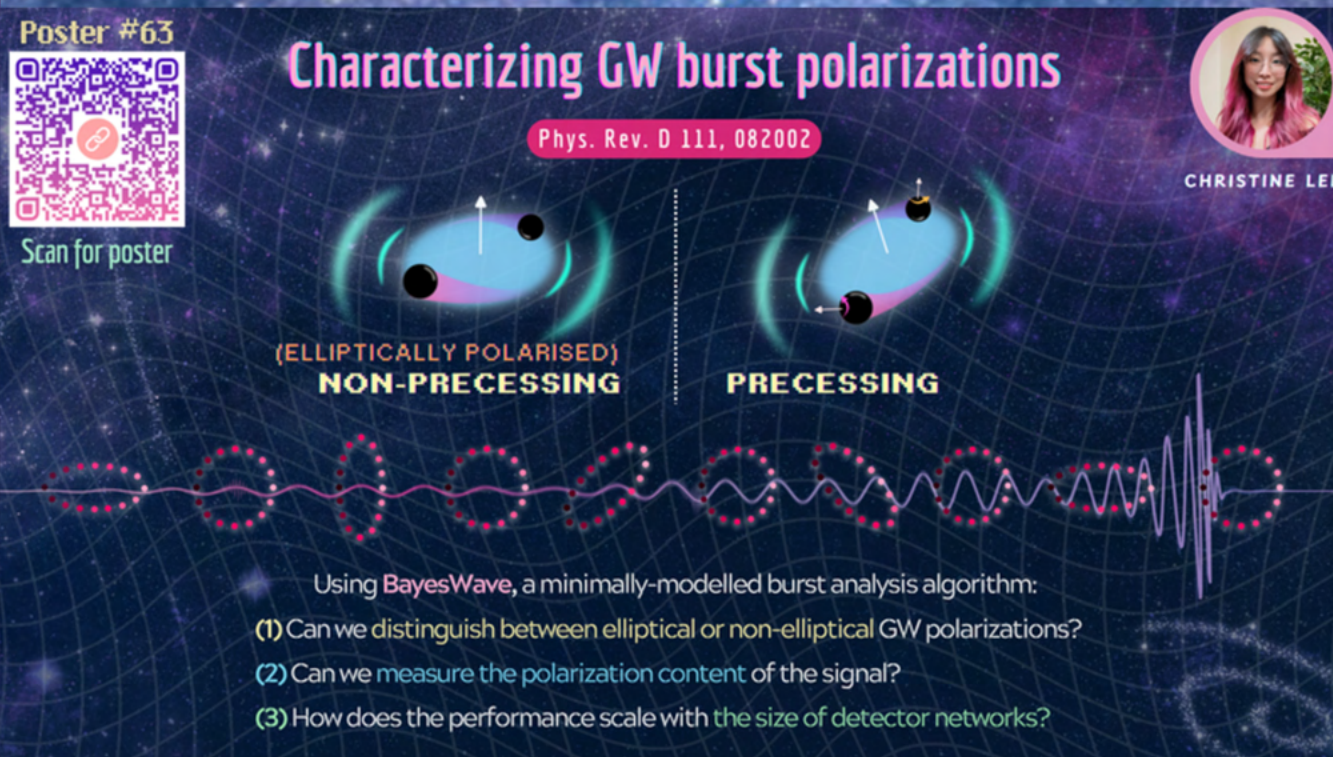


🌟 Sparkler slides 🌟

Title + photo
Get straight to the point
Keep it CLEAN and SHORT!!



(..and stay on brand!)



Characterization of gravitational-wave burst polarizations with the BayesWave algorithm

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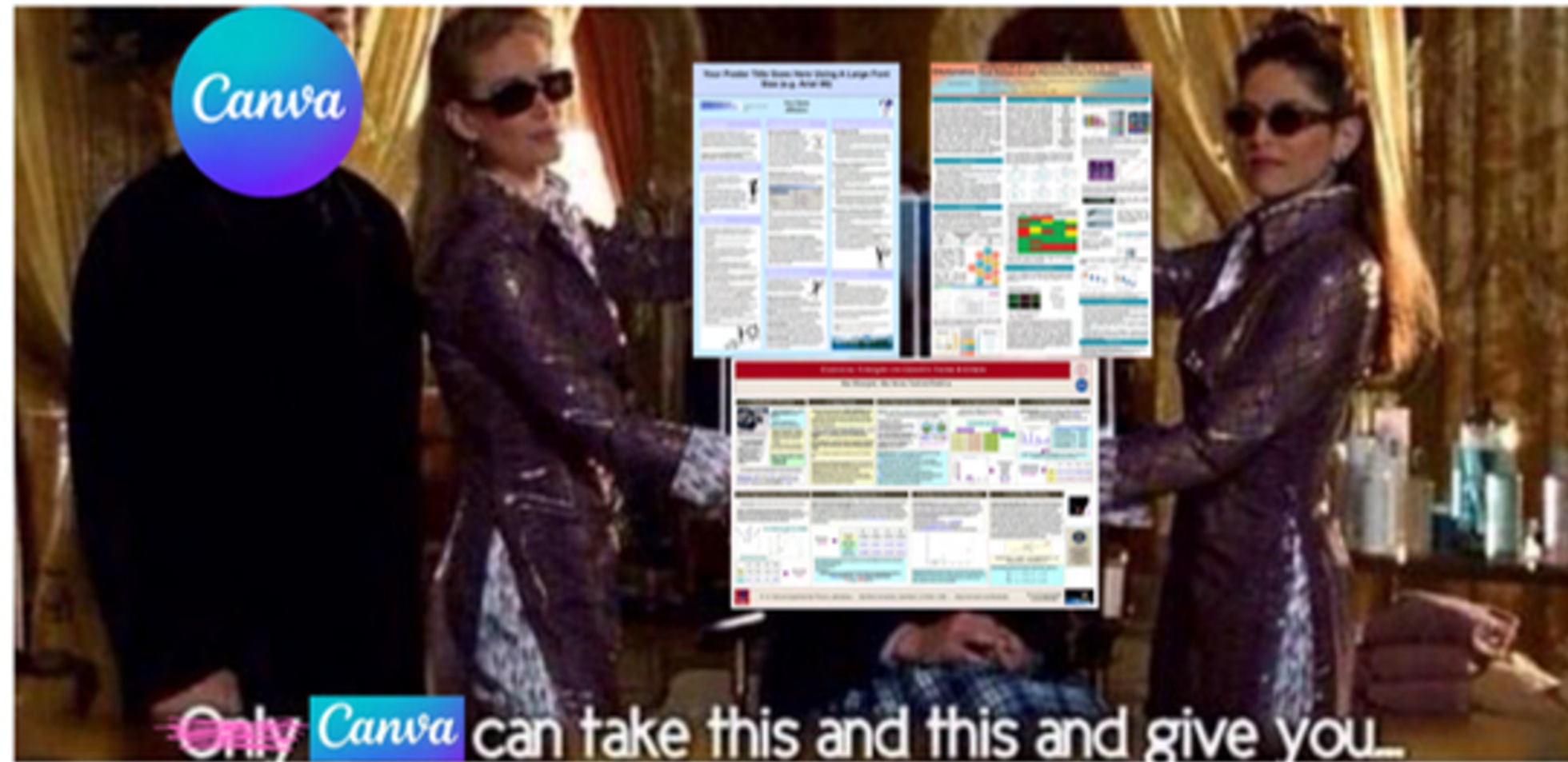
- ★ BayesWave is an unmodelled gravitational-wave (GW) burst characterization algorithm
- ★ How well can BayesWave recover polarization content of GW bursts?
- ★ Can we tell apart precessing and non-precessing compact binaries?
- ★ How does the performance improve with expanding detector networks?



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LIGO DCC G2400185



Designing your poster

with

Canva

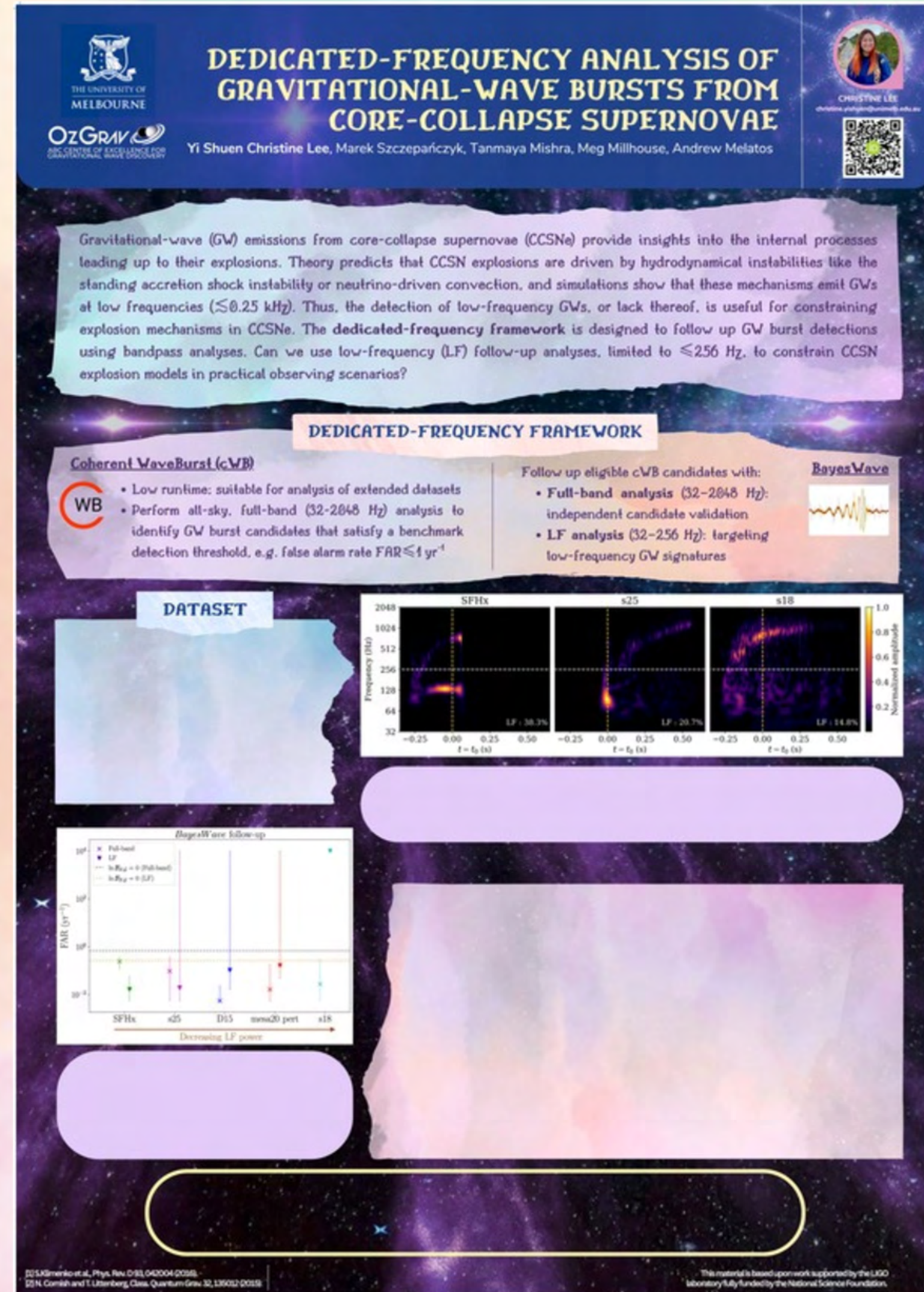
WHAT YOU NEED

- Canva account (FREE!)
- 400-word summary
- 2 plots/figures MAX
- A punchline

OPTIONAL

- A headshot
- ORCID QR Code

MAKING YOUR POSTER



1. **Create new design** - Choose the correct document size (you cannot change the size later with basic Canva).
2. **Find inspiration** from existing templates
3. **Create title** - Include authors, arXiv/article number (if available) and your contact details + your photo and ORCID (recommended).
4. **Plan layout** - Roughly section your poster
5. **Add content** - Add in your materials, highlight your punchline, and refine

EXTRA POINTERS

- **Avoid clutter** - Keep text brief and focused
- **Know your audience** - Highlight what you think interest them most
- **Use visuals** - Add bright colors and fun graphics to stand out (make full use of free Canva resources!!)
- **Readability and flow** - Make sure the text is legible from a distance and the content flows logically.



Canva

Suggestions?



Poster design with Canva



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